Let M be a smooth manifold

A map $f: M \to \mathbb{R}$ is smooth if for any coord. map σ on $U \subset \mathbb{R}^n$, $f \circ \sigma$ is C^∞

$$C^\infty(M) = \{\text{smooth functions } M \to \mathbb{R}\}$$

(smoothness for maps between mflds is defined the same way, in local coord.'s)

Tangent vectors

def: A tangent vector $X \in T_pM$ is a directional derivative

$$X: C^\infty(M) \to \mathbb{R}$$

such that

$$X(fg) = f(p) X_g + g(p) X_f$$

As in the surface case, T_pM is and n-dim. vector space

In local coord.'s $(x^1, \dots, x^n)$, the coordinate basis for T_pM is given by

$$\partial_i := \frac{\partial}{\partial x^i} \qquad i = 1, \dots, n$$

For a curve $\gamma: I \to M$, the velocity is

$$\dot{\gamma}: f \mapsto \frac{d}{dt}(f \circ \gamma)$$

The space $TM = \bigcup_{p \in M} T_pM$ has the structure of a manifold

Given a coordinate patch $\sigma : U \to M$ we define

$$\tilde{\sigma} : U \times \mathbb{R}^n \longrightarrow TM$$

by $$(x, v) \longmapsto v^j \partial_j \in T_{\sigma(x)} M$$

TM is called the tangent bundle of M

def.: A vector bundle E of rank k over M is a mfld. E equipped with a smooth map

$$\pi : E \to M$$

such that:

1) The fiber $E_p := \pi^{-1}(p)$ is a k-dim. vector space
2) For each $p \in M$, $\exists$ a nbhd. $V \ni p$ a diffeomorphism $\psi : \pi^{-1}(V) \longrightarrow V \times \mathbb{R}^k$

A section of E is a smooth map $f : M \to E$ such that $f(p) \in E_p$ for each p

i.e. $\pi \circ f = \text{identity map}$

A section of TM is called a vector field

Cotangent vectors

For a finite dim. vector space V, the dual space V^* is the set of linear maps $V \to \mathbb{R}$

The <u>cotangent space</u> T_p^*M = dual space of T_pM

Given coordinates x^i near p, we have the coord. basis ∂_j for T_pM

$\Rightarrow$ dual basis dx^i defined by $dx^i(\partial_j) = \delta^i_j$

This allows us to define coord. patches for $T^*M := \bigcup_p T_p^*M$

$\Rightarrow$ <u>cotangent bundle</u> T^*M

The <u>differential</u> of a smooth function $f \in C^\infty(M)$ defines a section of T^*M

$$df(X) := Xf \quad \text{for } X \in T_pM$$

This is consistent with the cotangent basis notation

$$dx^i(\partial_k) := \frac{\partial}{\partial x^k}(x^i) = \delta^i_k$$

Riemannian metrics

A metric on M is a smooth family of inner products

$$g(\cdot,\cdot) : T_pM \times T_pM \to \mathbb{R}$$

In local coord.'s this means that $g_{ij} := g(\partial_i, \partial_j)$ are smooth functions of $x^1, \dots x^n$ and $[g_{ij}]$ is symmetric & pos def at each pt.

The metric defines lengths of curves $\gamma : I \to M$

$$\ell(\gamma) = \int_I |\dot\gamma|_g \, dt \quad \text{where } |\dot\gamma|_g := \sqrt{g(\dot\gamma,\dot\gamma)}$$

g defines a map $T_pM \to T_p^*M$ by
$X \mapsto g(X, \cdot)$

In coord's, $\partial_i \mapsto g(\partial_i, \cdot) = g_{ij}\, dx^j$

Similarly, we can map $g^{-1}: T_p^*M \to T_pM$ using the inverse matrix

$$[g^{ij}] := [g_{ij}]^{-1}$$

Example: the gradient ∇f is defined by $g^{-1}(df)$

This means that $g(\nabla f, \cdot) = df(\cdot)$

In coord's $\quad df = (\partial_j f)\, dx^j$

$$\nabla f = g^{ij}(\partial_j f)\, \partial_i$$

Tensors

In linear algebra the tensor product of vector spaces V, W is a vector space $V \otimes W$ consisting of elements $v \otimes w$ with identifications:

1) $\lambda(v \otimes w) = (\lambda v) \otimes w = v \otimes (\lambda w)$ for $\lambda \in \mathbb{R}$
2) $(v_1 + v_2) \otimes w = v_1 \otimes w + v_2 \otimes w$, $\quad v \otimes (w_1 + w_2) = v \otimes w_1 + v \otimes w_2$

note that $\dim(V \otimes W) = \dim V \cdot \dim W$

example: $V^* \otimes V^* = \{\text{bilinear forms } V \times V \to \mathbb{R}\}$

Given vector bundles E, F over M, we can take the tensor products of fibers to define

$$E \otimes F := \bigcup_{p \in M} (E_p \otimes F_p)$$

Define: A tensor on M is a section of the bundle

$$T^k_l M := \underbrace{T^*M \otimes \cdots T^*M}_{k} \otimes \underbrace{TM \otimes \cdots \otimes TM}_{l}$$

example: g is a symmetric $(2,0)$ tensor